CHAPTER 7: PHYSICAL OPTICS

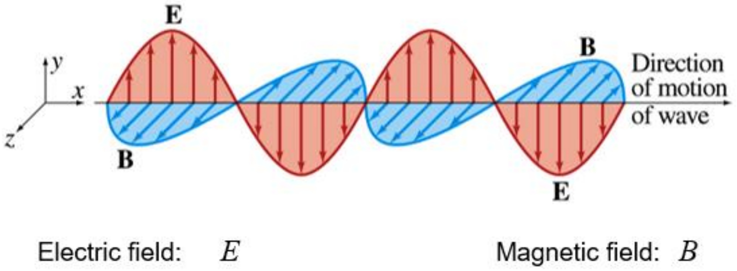
At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES
7.1	Constructive and destructive interferences a) Define interference of light. b) Define coherence. c) State the conditions of constructive and destructive interferences for in phase and anti-phase sources. **emphasise on the path difference and its equivalent to phase difference*.
7.2	Interference of transmitted light through double-slit a) Use i. for bright fringes (maxima), $y_m = \frac{m\lambda D}{d}$ ii. for dark fringes (minima), $y_m = \frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$ where m $= \pm 1,\ \pm 2,\ \pm 3, \ldots$ **Bright fringes:* *m= 0, central or 0th order max* *m= 1, first bright or 1st order max* *Dark fringes:* *m= 0, first dark or 0th order min* *m= 1, 2nd dark or 1st order min* b) Use $\Delta y = \frac{\lambda D}{d}$ and explain the effect of changing any of the variables. *Δy*: separation between two consecutive dark or bright fringe.*

7.1 Constructive and Destructive Interferences

Interference of Light

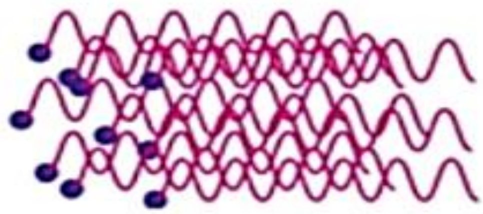
1. Light wave is an electromagnet waves (*emw*).
2. It consists of **varying electric field *E*** and **varying magnetic field *B*** which are **perpendicular to each other** as shown in figure below.



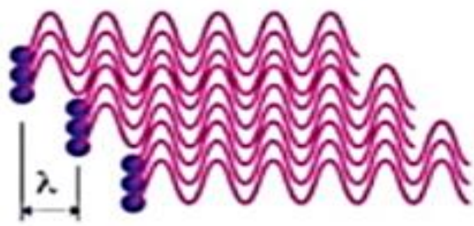
3. **Interference** is defined as **the effect of interaction between two or more waves which overlaps or superposed at a point and at a particular time from the sources**.
4. **Constructive interference** is defined as a reinforcement of amplitudes of light waves that will produce a bright fringe (maximum).
5. **Destructive interference** is defined as a total cancellation of amplitudes of light waves that will produce a dark fringe (minimum).

Coherence

1. The sources must have the **same wavelength and frequency** (monochromatic).
2. The sources must have a **constant phase difference** between them.

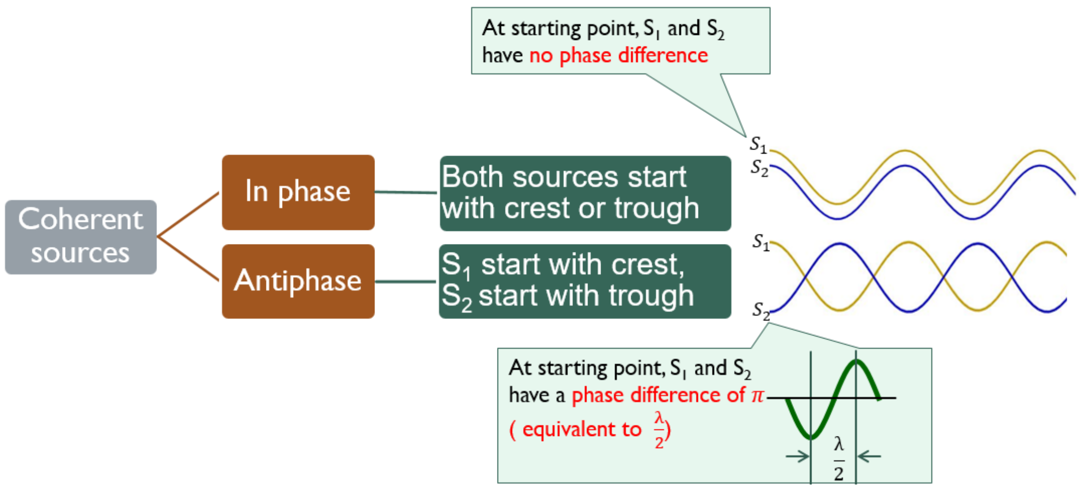


Incoherent waves

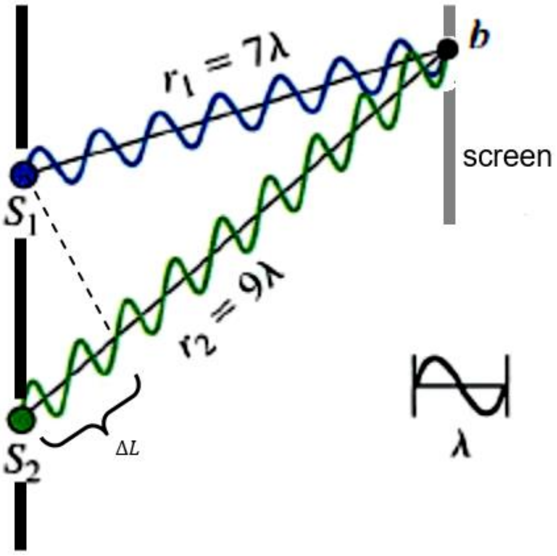


Coherent waves

3. There are two types of coherent sources:



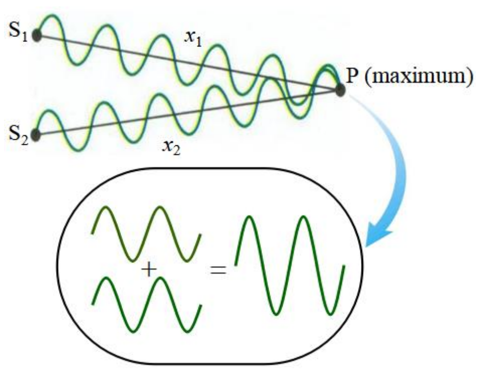
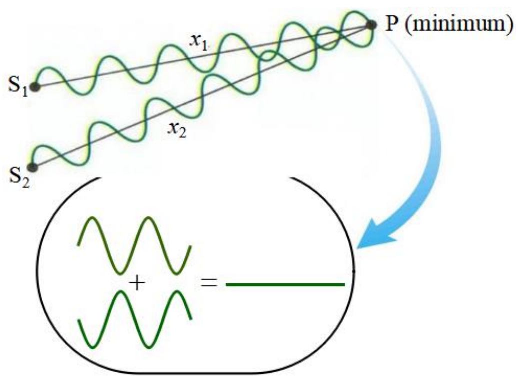
Path Difference and Phase Difference



Path difference, $\Delta \boldsymbol{L} = |S_2b - S_1b|$
$= |r_2 - r_1|$
$= |9\boldsymbol{\lambda} - 7\boldsymbol{\lambda}|$
$= 2\boldsymbol{\lambda}$

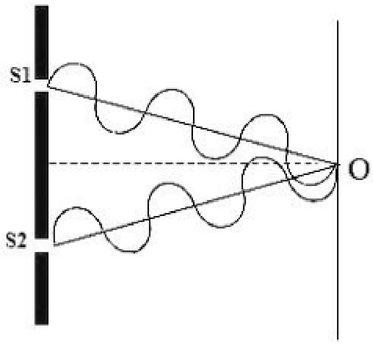
1. Path difference ($\Delta \boldsymbol{L}$) is defined as **difference in distances traveled** by two light waves **meeting at a point.** Phase difference is proportional to the difference in path length from the two sources to point ***b***.
2. It has a **direct equivalence to phase difference.** When path difference is one wavelength, λ the phase difference is one cycle ($\phi = 2\pi\ rad = 360^0$).
3. The path difference **decides whether constructive or destructive interference occurs** when they arrive at a point.

Conditions for Constructive and Destructive Interference

Path difference for constructive interference S_1 and S_2 are two coherent sources in phase	**Path difference for destructive interference** S_1 and S_2 are two coherent sources in phase
	

Conditions for Constructive Interference

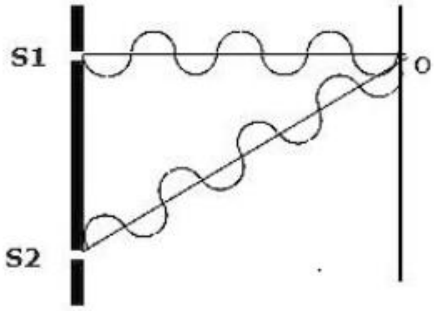
Two Coherence sources in-phase



Path Difference

$$\Delta L = [S_2O - S_1O]$$
$$= 0, \lambda, 2\lambda$$
$$= m\lambda$$

Two Coherence sources anti-phase



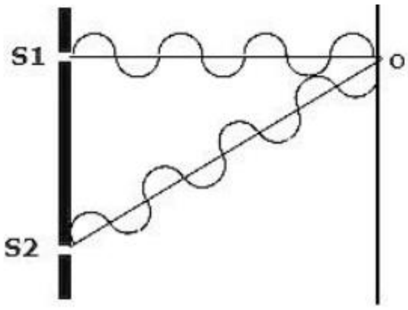
Path Difference

$$\Delta L = [S_2O - S_1O]$$
$$= \frac{\lambda}{2}, \frac{3\lambda}{2}, \lambda$$
$$= \left(m + \frac{1}{2}\right)\lambda$$

Two light waves arrive at O are in phase and give rise to constructive interference. A bright light is observed at O.

Conditions for Destructive Interference

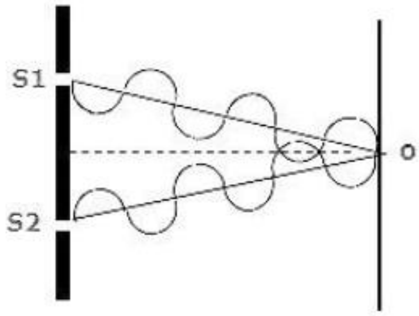
Two Coherence sources in-phase



Path Difference

$$\Delta L = [S_2O - S_1O]$$
$$= \frac{\lambda}{2}, \frac{3\lambda}{2}, \lambda$$
$$= \left(m + \frac{1}{2}\right)\lambda$$

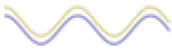
Two Coherence sources anti-phase



Path Difference

$$\Delta L = [S_2O - S_1O]$$
$$= 0, \lambda, 2\lambda$$
$$= m\lambda$$

Two light waves arrive at O are out of phase and give rise to destructive interference. A darkness is observed at O.

Two Coherent sources	Conditions	
	Constructive interference {Bright or Maxima}	**Destructive** interference {Dark or Minima}
In phase 	$\Delta L = m\lambda$ $m = 0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots$ $\Delta L = 0, \lambda, 2\lambda, 3\lambda, 4\lambda \cdots$	$\Delta L = (m + ½)\lambda$ $m = 0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots$ $\Delta L = 0.5\lambda, 1.5\lambda, 2.5\lambda, 3.5\lambda, 4.5\lambda \cdots$
Antiphase 	$\Delta L = (m + ½)\lambda$ $m = 0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots$ $\Delta L = 0.5\lambda, 1.5\lambda, 2.5\lambda, 3.5\lambda, 4.5\lambda \cdots$	$\Delta L = m\lambda$ $m = 0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots$ $\Delta L = 0, \lambda, 2\lambda, 3\lambda, 4\lambda \cdots$

Example 1

Two point sources X and Y emit waves of wavelength 2.0 cm in phase. The point P is 6.0 cm from X and 10.0 cm from Y. Another point Q is 7.0 cm from X and 4.0 cm from Y.

a) Determine the path difference of waves from X and Y at point P & point Q.

b) Deduce whether constructive interference or destructive interference occurs at P and Q.

Answers:

a) $\lambda = 2.0$ cm

For point P,

Path difference, $\Delta L = 10.0 \text{ cm} - 6.0 \text{ cm}$

$= 4.0$ cm

$= 2\,\lambda$

For point Q,

Path difference, $\Delta L = 7.0 \text{ cm} - 4.0 \text{ cm}$

$= 3.0$ cm

$= \frac{3}{2}\lambda$

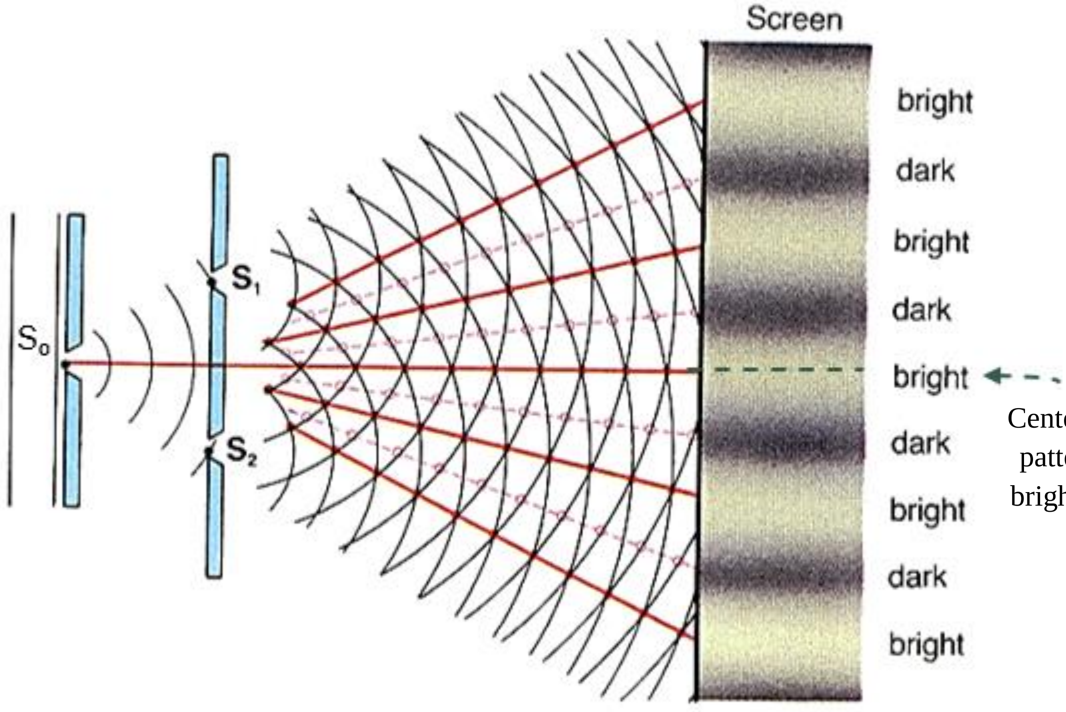
b) Both sources are in phase.

At point P, $\Delta L = 2\,\lambda$. So constructive interference occurs.

At point Q, $\Delta L = \frac{3}{2}\lambda$. So destructive interference occurs.

7.2 Interference of Transmitted Light Through Double-Slits

1. A monochromatic light source is placed behind a narrow slit S_o, which acts as point source.



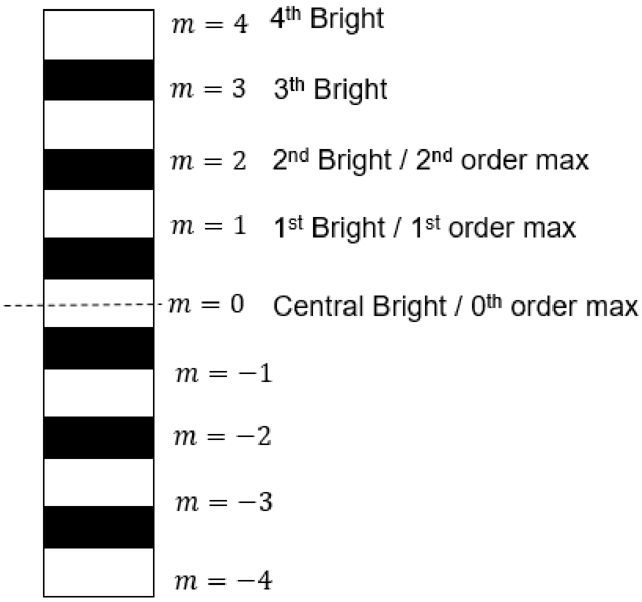
Center of the pattern is a bright fringe

2. Cylindrical wave fronts spread out from slit S_o and **reach S_1 and S_2 in phase** because both slits are **equal distances from S_o**.
3. The waves **emerging from slits S_1 and S_2** are therefore also **always in phase**.
4. So S_1 & S_2 are **coherent sources**.
5. Waves spread out behind each slits.
6. The waves **interfere** in the region where they overlap, resulting in a pattern of bright and dark bands on the screen.
7. **Bright** fringe is observed when **constructive** interference occurs and **intensity is maximum**.
8. **Dark** fringe is observed when **destructive** interference occurs and **intensity is minimum**.
9. **Center** of the pattern is a **bright** fringe.

Distance of m^{th} bright fringe (maxima) from central bright :

$$y_m = \frac{m\lambda D}{d}$$

Where,
λ : wavelength of the light
d : separation between 2 slits
D : Distance from slits to screen
y_m : Distance of m^{th} from central
m : order number
$(0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots)$

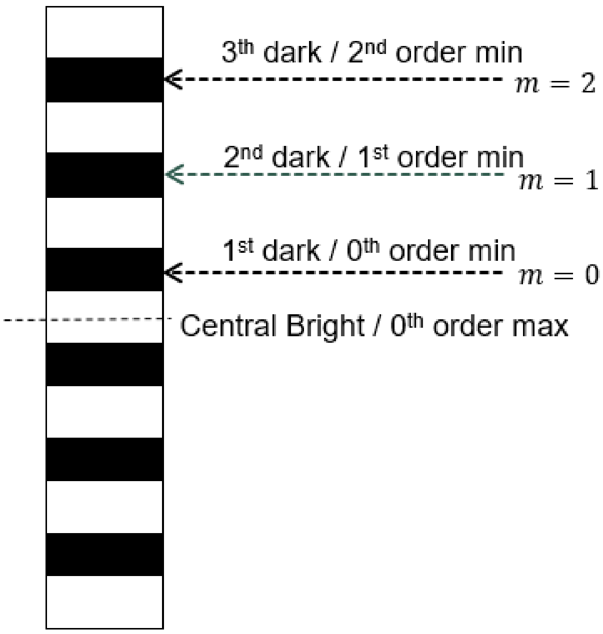


Interference pattern

Distance of m^{th} dark fringe (minima) from central bright :

$$y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$$

Where,
λ : wavelength of the light
d : separation between 2 slits
D : Distance from slits to screen
y_m : Distance of m^{th} from central
m : order number
$(0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots)$



Interference pattern

Example 2

An interference pattern is formed on a screen when light of wavelength 550 nm is incident on two parallel slits 50μm apart. The second-order bright fringe is 4.5 cm from the center of the central maximum. How far from the slits is the screen?

Answers:

$\lambda = 550\ nm = 550 \times 10^{-9} m$

$d = 50\mu m = 50 \times 10^{-6} m$

$m = 2$

$y_2 = 4.5\ cm = 4.5 \times 10^{-2} m$

$y_m = \frac{m\lambda D}{d}$

$y_2 = \frac{2\lambda D}{d}$

$4.5 \times 10^{-2} = \frac{(2)(550 \times 10^{-9})D}{50 \times 10^{-6}}$

$D = 2.05\ m$

Example 3

In a lab experiment, monochromatic light passes through two narrow slits that are 0.05 mm apart. The interference pattern is observed on a white wall 1.0 m from the slits, and the second-order bright is 2.4 cm from the center of the central maximum.

a) Determine the wavelength of the light.

b) Calculate the distance between the second-order and third-order bright fringes?

Answers:

$d = 0.05\ mm = 0.05 \times 10^{-3} m$

$D = 1.0\ m$

$y_2 = 2.4 \times 10^{-2} m$

a) $y_m = \frac{m\lambda D}{d}$

$y_2 = \frac{2\lambda D}{d}$

$2.4 \times 10^{-2} = \frac{(2)(\lambda)(1.0)}{0.05 \times 10^{-3}}$

$\lambda = 6.0 \times 10^{-7} m$

b) $x = y_3 - y_2$

$x = \frac{3\lambda D}{d} - \frac{2\lambda D}{d}$

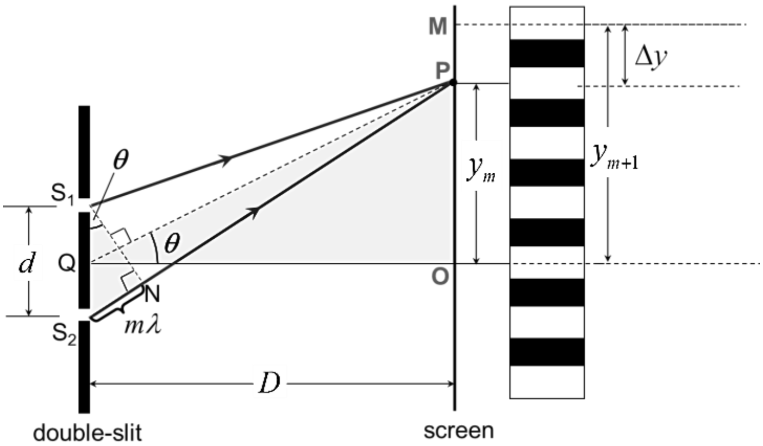
$x = \frac{\lambda D}{d}$

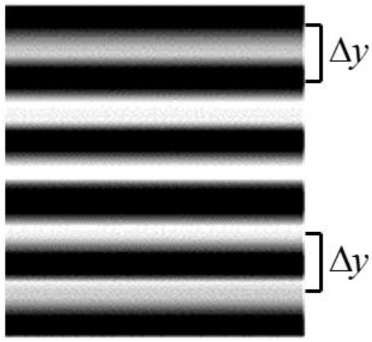
$x = \frac{(6.0 \times 10^{-7})(1.0)}{0.05 \times 10^{-3}}$

$x = 1.2 \times 10^{-2} m$

Separation between two consecutive bright or dark fringes, Δy

1. Also known as fringe width or fringe separation.





$$\Delta y = \frac{\lambda D}{d}$$

2. From the equation Δy depends on wavelength, distance apart, d of double slits and the distance between slits and the screen.
3. What are the effects of changing $\boldsymbol{\lambda}$, D or d on Δy (or interference pattern)?
   Simulation:

Refer : $\Delta y = \dfrac{\lambda D}{d}$ 

$$\rightarrow \Delta y \propto \boldsymbol{\lambda}\ ;\ \Delta y \propto D\ ;\ \Delta y \propto \frac{1}{d}$$

Variables	Δy	Interference pattern	
Increasing D (move screen farther from slits)	Δy ↑	Fringes will be farther apart	
Shorter λ	Δy ↓	Fringes will be closer together	
Decreasing d (slits move closer together)	Δy ↑	Fringes will be farther apart	

Example 4

Figure shows two coherent sources of light in phase. The separation of S_1 and S_2 is 1.2 mm and the screen is 2.5 m from the sources. If the wavelength of the light used is 520 nm, calculate the separation between two consecutive bright fringes.

Answer:

$d = 1.2\ mm = 1.2 \times 10^{-3} m$

$D = 2.5\ m$

$\lambda = 520\ nm = 520 \times 10^{-9} m$

$$\Delta y = \frac{\lambda D}{d}$$

$$\Delta y = \frac{(520 \times 10^{-9})(2.5)}{1.2 \times 10^{-3}}$$

$\Delta y = 1.08 \times 10^{-3} m$

Example 5

In a Young's double slit experiment, the slit separation is 0.20 mm and a screen is 0.40 m from the double-slit. A light source that emits red and violet light of wavelengths 700 nm and 420 nm is used. Calculate the separation between red fringes and violet fringes on the screen.

Answers:

$d = 0.2\ mm = 0.2 \times 10^{-3} m$

$D = 0.40\ m$

$\lambda_{red} = 700\ nm = 700 \times 10^{-9} m$

$\lambda_{violet} = 420\ nm = 420 \times 10^{-9} m$

$$\Delta y = \frac{\lambda D}{d}$$

$$\Delta y_{red} = \frac{(700 \times 10^{-9})(0.40)}{0.2 \times 10^{-3}} = 1.4\ mm$$

$$\Delta y_{violet} = \frac{(420 \times 10^{-9})(0.40)}{0.2 \times 10^{-3}} = 0.84 \text{ mm}$$

SELECTED FORMULAE

1. $y_m = \frac{m\lambda D}{d}$
2. $y_m = \frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$
3. $\Delta y = \frac{\lambda D}{d}$

DIY Quiz

Instruction:
Answer all questions in 10 minutes.
No discussion & do not refer to any notes.
You can only refer to the formulae given below.

1. In a double slit experiment, the light source simultaneously emits blue light of wavelength 400 nm and yellow light of wavelength 600 nm. The slits are 0.08 mm apart and the interference pattern is observed on a screen at a distance of 60.0 cm from the slits. Calculate
   a) The distances of the first blue and the first yellow fringes from the central fringe (that is, the zeroth fringe).
   b) The shortest distance from the central fringe where the yellow and blue fringes overlap.

   Answers:
   a) $y_{blue} = 3.0\ mm$; $y_{yellow} = 4.5\ mm$
   b) $y = 9\ mm$

<u>PAST YEAR QUESTIONS</u>

A red light of wavelength 680 nm incident on the double slit with separation distance 0.1 mm. The screen is placed at 1.0 m from slit.

a) Calculate the separation distance between two consecutive bright fringes.

b) If the red light is replaced with a violet light ($\lambda = 420nm$), state the effect of the fringe separation. Explain your answer.

c) If separation distance of the double slits is decreased, state the effect to the separation distance between two consecutive dark fringes. Explain your answer.

[6 marks]